

1. $y'' + y = 0$, $y(0) = 0$, $y(\pi) = 1$

Solⁿ:- $y'' + y = (D^2 + 1)y = 0$

Auxiliary equation is $m^2 + 1 = 0$

$$\Rightarrow m^2 = -1$$

$$\Rightarrow m = \pm i$$

$\Rightarrow y = C_1 \cos x + C_2 \sin x$ is the auxiliary solution

To find C_1 and C_2

$$y(0) = 0 \Rightarrow y(0) = C_1 \cos 0 + C_2 \sin 0$$

$$\Rightarrow 0 = C_1 + 0$$

$$\Rightarrow C_1 = 0$$

$$y(\pi) = 1 \Rightarrow y(\pi) = C_1 \cos \pi + C_2 \sin \pi$$

$$\Rightarrow 1 = C_1(-1) + C_2(0)$$

$$\Rightarrow -C_1 = 1 \Rightarrow C_1 = -1$$

For C_1 we got 0 and -1 value, which is not possible.

Hence the BVP has no solution.

2. $y'' + y = 0$, $y(0) = 0$ and $y(\pi) = 0$

Soln:- $y = c_1 \cos x + c_2 \sin x$

$$y(0) = 0 \Rightarrow 0 = c_1 \cos 0 + c_2 \sin 0 \\ \Rightarrow c_1 = 0$$

$$y(\pi) = 0 \Rightarrow 0 = c_1 \cos \pi + c_2 \sin \pi \\ \Rightarrow -c_1 = 0 \Rightarrow c_1 = 0$$

Here $c_1 = 0$ and c_2 is a free variable.
 $\therefore$ This BVP has infinitely many solutions.

Let $c_2 = B$, be some arbitrary scalar.
Then,

$$y = B \sin x \text{ is the solution}$$

3. $y'' + y = 0$, $y(0) = 0$, $y(\frac{\pi}{2}) = 1$

Soln:- $y = c_1 \cos x + c_2 \sin x$

$$y(0) = 0 \Rightarrow c_1 = 0$$

$$y(\frac{\pi}{2}) = 1 \Rightarrow 1 = c_1 \cos \frac{\pi}{2} + c_2 \sin \frac{\pi}{2} \\ \Rightarrow 1 = 0 + c_2 (1) \\ \Rightarrow c_2 = 1$$

As we get unique solution for c_1 and c_2 ,

the BVP has unique solution and the solution is

$$y = \sin x.$$

Solution of BVP - Finite difference method.

By Taylor series expansion of $y(x)$, we have

$$y_{i+1} = y(x_i + h) = y_i + h y_i' + \frac{h^2}{2!} y_i'' + \frac{h^3}{3!} y_i''' + \dots$$

$\hookrightarrow \textcircled{1}$

$$y_{i-1} = y(x_i - h) = y_i - h y_i' + \frac{h^2}{2!} y_i'' - \frac{h^3}{3!} y_i''' + \dots$$

$\hookrightarrow \textcircled{2}$

Subtracting $\textcircled{2}$ from $\textcircled{1}$,

$$y_{i+1} - y_{i-1} = 2 \left[h y_i' + \frac{h^3}{3!} y_i''' + \frac{h^5}{5!} y_i^{(v)} + \dots \right]$$

$$\Rightarrow \boxed{y_i' = \frac{y_{i+1} - y_{i-1}}{2h}} \quad \text{---} \textcircled{3}$$

with an error $O(h^2)$ (as h is small, neglect $h^3, h^5, \dots$)

Also adding eqns ① and ② we get

$$\begin{aligned} y_{i+1} + y_{i-1} &= 2y_i + 2 \frac{h^2}{2!} y_i'' + 2 \frac{h^4}{4!} y_i^{(iv)} + \dots \\ &= 2y_i + h^2 y_i'' + \frac{h^4}{12} y_i^{(iv)} + \dots \end{aligned}$$

$$\Rightarrow \boxed{y_i'' = \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}} \quad \text{--- ④}$$

Now consider the finite difference method of solving linear BVP of the form

$$p(x) y'' + q(x) y' + r(x) y = f(x) \quad \text{--- ⑤}$$

We consider two types of boundary conditions

Type 1) The boundary conditions do not involve y' .

Type 2) The boundary conditions involve y' .

I) The boundary conditions do not involve y'

Consider (by setting $x=x_i$ in $\textcircled{*}$)

$$p_i y_i'' + q_i y_i' + r_i y_i = f_i \quad \text{--- } \textcircled{a}$$

$$p_i = p(x_i), \quad q_i = q(x_i), \quad r_i = r(x_i), \quad f_i = f(x_i)$$

$$y_i' = y'(x_i), \quad y_i'' = y''(x_i), \quad y_i = y(x_i)$$

$$\text{Let } y(a) = y_a, \quad y(b) = y_b \quad \text{--- } \textcircled{b}$$

Replace y_i' and y_i'' in $\textcircled{a}$ by $\textcircled{3}$ & $\textcircled{4}$

$$\Rightarrow p_i \left[\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} \right] + q_i \left[\frac{y_{i+1} - y_{i-1}}{2h} \right] + r_i y_i = f_i \quad \text{--- } \textcircled{c}$$

$$i = 1, 2, \dots, n-1$$

The eqn $\textcircled{c}$ gives rise to system of equations of order $n-1$.

$$\begin{array}{ccccccc} x_0 & x_1 & x_2 & x_3 & \dots & x_n & \\ | & | & | & | & & | & \\ \hline x=a & & & & & x=b & \end{array} \quad \begin{array}{l} x_1 = x_0 + h \\ x_2 = x_1 + h \\ \vdots \end{array}$$

$$\text{Given } y(x_0) = y_0 = y(a) = y_a$$

$$y(x_n) = y_n = y(b) = y_b$$

Our aim is to find $x_1, x_2, x_3, \dots$

1. Solve $xy'' + y = 0$, $y(1) = 1$, $y(2) = 2$
with $h = 0.5$ and $h = 0.25$

Soln :-

a) $h = 0.5$

Here $x_0 = 1 = a$

$$x_1 = x_0 + h = 1.5$$

$$x_2 = x_1 + h = 2 = b$$

$$\begin{array}{ccccc} & | & & | & & | \\ & x_0 = 1 & & x_1 = 1.5 & & x_2 = 2 \\ y_0 = y(1) = 1 & & & \downarrow & & y_2 = y(2) = 2 \\ & & & y_1 = y(1.5) = ? & & \end{array}$$

Consider $xy'' + y = 0$

Replace x by x_i y by y_i

$$x_i y_i'' + y_i = 0 \quad \text{--- (1)}$$

$$x_i \left[\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} \right] + y_i = 0$$

$$\text{Here, } h = 0.5 = \frac{1}{2} \Rightarrow h^2 = \frac{1}{4}$$

$$\Rightarrow x_i \left[\frac{y_{i+1} - 2y_i + y_{i-1}}{1/4} \right] + y_i = 0$$

$$\Rightarrow 4x_i (y_{i+1} - 2y_i + y_{i-1}) + y_i = 0$$

$$\Rightarrow 4x_i y_{i+1} - 8x_i y_i + 4x_i y_{i-1} + y_i = 0 \quad \text{--- (2)}$$

$$\text{Put } i=1$$

$$\Rightarrow 4x_1 y_2 - 8x_1 y_1 + 4x_1 y_0 + y_1 = 0$$

$$\Rightarrow 4(1.5)(2) - 8(1.5)y_1 + 4(1.5)(1) + y_1 = 0$$

$$\Rightarrow 12 - 12y_1 + 6 + y_1 = 0$$

$$\Rightarrow -11y_1 + 18 = 0$$

$$\Rightarrow y_1 = \frac{18}{11} = 1.6364$$

$$\Rightarrow y(1.5) = 1.6364$$

$$b) \quad h = 0.25$$

$$x_0 = 1, \quad x_1 = 1.25, \quad x_2 = 1.5, \quad x_3 = 1.75, \quad x_4 = 2$$

$$y_0 = y(1) \quad y_1 = ? \quad y_2 = ? \quad y_3 = ? \quad y_4 = y(2)$$

$$= 1 \quad \quad \quad \quad \quad = 2$$

$$\begin{array}{ccccccccc} & | & & | & & | & & | & & | \\ x_0 & & x_1 & & x_2 & & x_3 & & x_4 \end{array}$$

$$y_0 = 1 \quad y_1 = ? \quad y_2 = ? \quad y_3 = ? \quad y_4 = 2$$

$$x_i y_i'' + y_i = 0$$

$$x_i \left[\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} \right] + y_i = 0$$

$$\text{Put } h = 0.25 = 1/4 \Rightarrow h^2 = 1/16$$

$$\Rightarrow 16x_i (y_{i+1} - 2y_i + y_{i-1}) + y_i = 0$$

$$\Rightarrow 16x_i y_{i+1} - 32x_i y_i + 16x_i y_{i-1} + y_i = 0$$

$$\text{Put } i=1 \Rightarrow 16x_1 y_2 - 32x_1 y_1 + 16x_1 y_0 + y_1 = 0$$

$$\Rightarrow 16(1.25)y_2 - 32(1.25)y_1 + 16(1.25)(1) + y_1 = 0$$

$$\Rightarrow 20y_2 - 40y_1 + 20 + y_1 = 0$$

$$\Rightarrow -39y_1 + 20y_2 = -20 \quad \text{--- (1)}$$

$$\text{Put } i=2 \Rightarrow 16x_2 y_3 - 32x_2 y_2 + 16x_2 y_1 + y_2 = 0$$

$$\Rightarrow 16(1.5)y_3 - 32(1.5)y_2 + 16(1.5)y_1 + y_2 = 0$$

$$\Rightarrow 24y_3 - 48y_2 + 24y_1 + y_2 = 0$$

$$\Rightarrow 24y_1 - 47y_2 + 24y_3 = 0 \quad \text{--- (2)}$$

$$\text{Put } i=3 \Rightarrow 16x_3y_4 - 32x_3y_3 + 16x_3y_2 + y_3 = 0$$

$$\Rightarrow 16(1.75)(2) - 32(1.75)y_3 + 16(1.75)y_2 + y_3 = 0$$

$$\Rightarrow 56 - 56y_3 + 28y_2 + y_3 = 0$$

$$\Rightarrow 28y_2 - 55y_3 = -56 \quad \text{--- (3)}$$

Consider (1), (2) and (3)

$$-39y_1 + 20y_2 = -20 \quad \text{--- (1)}$$

$$24y_1 - 47y_2 + 24y_3 = 0 \quad \text{--- (2)}$$

$$28y_2 - 55y_3 = -56 \quad \text{--- (3)}$$

$$\text{(2)/24 \& (3)/55}$$

$$\Rightarrow y_1 - \frac{47}{24}y_2 + y_3 = 0$$

$$\frac{28}{55}y_2 - y_3 = \frac{-56}{55}$$

Add the above equations

$$y_1 - 1.4492y_2 = -1.0182 \quad \text{--- (4)}$$

$$\text{(1)/39} \Rightarrow -y_1 + \frac{20}{39}y_2 = \frac{-20}{39} \quad \text{--- (5)}$$

Add ④ & ⑤

$$-0.9364 y_2 = -1.531$$

$$\Rightarrow y_2 = 1.6349$$

$$-y_1 = -\left(\frac{20}{39}\right)(1.6349) - \frac{20}{39}$$

$$\Rightarrow y_1 = 1.3513$$

$$y_3 = \frac{28(1.6349) + 56}{55} = 1.8505$$

$$\Rightarrow y_3 = 1.8505$$

$$\therefore y_1 = y(1.25) = 1.3513$$

$$y_2 = y(1.5) = 1.6349$$

$$y_3 = y(1.75) = 1.8505$$

2. Solve $(x^3+1)y'' + x^2y' - 4xy = 2$,
 $y(0)=0$, $y(2)=4$ with $h=0.5$

Solⁿ:

$$(x_i^3+1)y_i'' + x_i^2 y_i' - 4x_i y_i = 2$$

$$\begin{array}{ccccccccc} y_0=0 & & y_1 & & y_2 & & y_3 & & y_4=4 \\ | & & | & & | & & | & & | \\ x_0=0 & & x_1=0.5 & & x_2=1 & & x_3=1.5 & & x_4=2 \end{array}$$

$$(x_i^3+1) \left[\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} \right] + x_i^2 \left[\frac{y_{i+1} - y_{i-1}}{2h} \right] - 4x_i y_i = 2$$

Put $h = 0.5 = \frac{1}{2} \Rightarrow h^2 = \frac{1}{4}$

$$\Rightarrow 4(x_i^3+1)(y_{i+1} - 2y_i + y_{i-1}) + x_i^2(y_{i+1} - y_{i-1}) - 4x_i y_i = 2$$

$$- 4x_i y_i = 2 \quad \text{--- } (*) \quad i=1, 2, 3$$

Put $i=1$

$$\Rightarrow 4(x_1^3+1)(y_2 - 2y_1 + y_0) + x_1^2(y_2 - y_0) - 4x_1 y_1 = 2$$

$$\Rightarrow 4(0.5^3+1)(y_2 - 2y_1 + 0) + 0.5^2(y_2 - 0) - 4(0.5)y_1 = 2$$

$$\Rightarrow (y_2 - 2y_1) + 0.25 y_2 - 2y_1 - 2 = 0$$

$$\Rightarrow 4.5 y_2 - 9 y_1 + 0.25 y_2 - 2y_1 - 2 = 0$$

$$\Rightarrow -11y_1 + 4.75 y_2 - 2 = 0$$

$$\Rightarrow -11y_1 + 4.75 y_2 = 2 \quad \text{--- } (1)$$

Put $i=2$

$$4(x_2^3+1)(y_3-2y_2+y_1)+x_2^2(y_3-y_1)-4x_2y_2=2$$

$$\Rightarrow 4(1+1)(y_3-2y_2+y_1)+(y_3-y_1)-4y_2=2$$

$$\Rightarrow 8y_3-16y_2+8y_1+y_3-y_1-4y_2=2$$

$$\Rightarrow 9y_3-20y_2+7y_1=2$$

$$\Rightarrow 7y_1-20y_2+9y_3=2 \quad \text{--- (2)}$$

Put $i=3$,

$$4(x_3^3+1)(y_4-2y_3+y_2)+x_3^2(y_4-y_2)-4x_3y_3=2$$

$$\Rightarrow 4(1.5^3+1)(4-2y_3+y_2)+1.5^2(4-y_2)-4 \times 1.5 \times y_3=2$$

$$\Rightarrow 17.5(4-2y_3+y_2)+2.25(4-y_2)-6y_3=2$$

$$\Rightarrow 70-35y_3+17.5y_2+9-2.25y_2-6y_3=2$$

$$\Rightarrow 15.25y_2-41y_3=-77 \quad \text{--- (3)}$$

$$\Rightarrow y_1 = y(x_1) = y(0.5) = 0.25$$

$$\Rightarrow y_2 = y(x_2) = y(1) = 1$$

$$\Rightarrow y_3 = y(x_3) = y(1.5) = 2.25$$

H.W

Solve $x^2 y'' + xy' + (x^2 - 3)y = 0$ with
 $y(1) = 0$, $y(2) = 2$ and $h = 0.25$

Ans:- 0.6044, 1.1304, 1.5973.

II) The boundary conditions involve y' .

Let the boundary conditions be

$$\alpha_0 y(a) + \alpha_1 y'(a) = \alpha_2$$

$$\beta_0 y(b) + \beta_1 y'(b) = \beta_2$$

Changing the derivatives into finite differences, we get

$$\alpha_0 y_0 + \alpha_1 y'_0 = \alpha_2$$

$$\beta_0 y_n + \beta_1 y'_n = \beta_2$$

$$y'_i = \frac{y_{i+1} - y_{i-1}}{2h} \Rightarrow y'_0 = \frac{y_1 - y_{-1}}{2h}$$

$$y'_n = \frac{y_{n+1} - y_{n-1}}{2h}$$

$$\therefore \alpha_0 y_0 + \alpha_1 \left[\frac{y_1 - y_{-1}}{2h} \right] = \alpha_2 \quad \text{--- (a)}$$

$$\beta_0 y_n + \beta_1 \left[\frac{y_{n+1} - y_{n-1}}{2h} \right] = \beta_2 \quad \text{--- (b)}$$

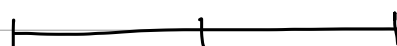
Here the problem involves finding y_{-1}, y_{n+1} (the values of y outside the interval of interest).

The eqn (a) will enable us to find y_1 in terms of y_0 and y_1 .

The eqn (b) will enable us to find y_{n+1} in terms of y_{n-1} and y_n .

1. Solve $xy'' + y = 0$, $y'(1) = 0$, $y(2) = 1$, $h = 0.5$

Soln:-

			$x_0 = 1$
$x_0 = 1$	$x_1 = 1.5$	$x_2 = 2$	$x_1 = 1 + 0.5 = 1.5$
$y_0 = ?$	$y_1 = ?$	$y_2 = y(2) = 1$	$x_2 = 1.5 + 0.5 = 2$

$$\text{Given } y'(1) = y'(x_0) = 0 \Rightarrow y'_0 = 0$$

$$\text{Consider } y_i' = \frac{y_{i+1} - y_{i-1}}{2h}$$

$$y'_0 = \frac{y_1 - y_{-1}}{2 \times \frac{1}{2}}$$

$$0 = \frac{y_1 - y_{-1}}{2}$$
$$\Rightarrow \boxed{y_1 = y_{-1}}$$

$$\text{Given } xy'' + y = 0$$

$$\Rightarrow x_i y_i'' + y_i = 0$$

$$x_i \left[\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} \right] + y_i = 0, \quad i=0,1$$

_____ *

Put $i=0$ in $\textcircled{*}$ and $h = \frac{1}{2}$

$$x_0 \left[\frac{y_1 - 2y_0 + y_{-1}}{\frac{1}{4}} \right] + y_0 = 0$$

$$4x_0 (y_1 - 2y_0 + y_{-1}) + y_0 = 0 \quad (\because y_{-1} = y_1)$$

$$\Rightarrow 4(1) (2y_1 - 2y_0) + y_0 = 0$$

$$\Rightarrow 8y_1 - 8y_0 + y_0 = 0$$

$$\Rightarrow 8y_1 - 7y_0 = 0 \quad \text{--- } \textcircled{a}$$

Put $i=1$ in $\textcircled{*}$ and $h = \frac{1}{2}$

$$x_1 \left[\frac{y_2 - 2y_1 + y_0}{\frac{1}{4}} \right] + y_1 = 0$$

$$\Rightarrow 4x_1 (y_2 - 2y_1 + y_0) + y_1 = 0$$

$$\Rightarrow 4(1.5) (1 - 2y_1 + y_0) + y_1 = 0$$

$$\Rightarrow 6 (1 - 2y_1 + y_0) + y_1 = 0$$

$$= -11y_1 + 6y_0 = -6 \quad \text{--- } \textcircled{b}$$

$$\Rightarrow y_0 = y(1) = \frac{48}{29} = 1.6552$$

$$y_1 = y(1.5) = \frac{42}{29} = 1.4483$$

2. Solve $y'' + (1+x)y' - y = 0$, $y(0) = y'(0)$,
 $y(1) + y'(1) = 1$ with $h = 0.5$

Soln:-

$y_0 = ?$	$y_1 = ?$	$y_2 = ?$
$x_0 = 0$	$x_1 = 0.5$	$x_2 = 1$

Given $y'(0) = y(0) \Rightarrow y'_0 = y_0$

Consider $y'_i = \frac{y_{i+1} - y_{i-1}}{2h}$

$$y'_0 = \frac{y_1 - y_{-1}}{2 \times \frac{1}{2}}$$

$$y_0 = y_1 - y_{-1} \quad (\because y'_0 = y_0)$$
$$\Rightarrow \boxed{y_{-1} = y_1 - y_0}$$

Given $y(1) + y'(1) = 1$

$$\Rightarrow y_2 + y'_2 = 1$$

But $y'_i = \frac{y_{i+1} - y_{i-1}}{2h}$

$$y'_2 = \frac{y_3 - y_1}{2 \times \frac{1}{2}} = y_3 - y_1$$

$$\Rightarrow y_2 + (y_3 - y_1) = 1$$

$$\Rightarrow \boxed{y_3 = 1 + y_1 - y_2}$$

Consider

$$y_i'' + (1+x_i) y_i' - y_i = 0$$

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + (1+x_i) \left[\frac{y_{i+1} - y_{i-1}}{2h} \right] - y_i = 0$$

$i=0, 1, 2$

$i=0$

$$\frac{y_1 - 2y_0 + y_{-1}}{1/4} + (1+x_0) \left[\frac{y_1 - y_{-1}}{2 \times 1/2} \right] - y_0 = 0$$

$$\Rightarrow 4(y_1 - 2y_0 + y_{-1}) + (1+0)(y_1 - y_{-1}) - y_0 = 0$$

$$\Rightarrow 4(y_1 - 2y_0 + y_1 - y_0) + (y_1 - y_1 + y_0) - y_0 = 0$$

$$\Rightarrow 8y_1 - 12y_0 = 0$$

$$\Rightarrow 2y_1 - 3y_0 = 0 \quad \text{--- (a)}$$

$i=1$

$$4(y_2 - 2y_1 + y_0) + (1+x_1)(y_2 - y_0) - y_1 = 0$$

$$\Rightarrow 4(y_2 - 2y_1 + y_0) + (1.5)(y_2 - y_0) - y_1 = 0$$

$$\Rightarrow 5.5y_2 - 9y_1 + 2.5y_0 = 0 \quad \text{--- (b)}$$

$$\underline{i=2}$$

$$4(y_3 - 2y_2 + y_1) + (1+1)(y_3 - y_1) - y_2 = 0$$

$$\Rightarrow 6y_3 - 9y_2 + 2y_1 = 0$$

$$\text{But } y_3 = 1 + y_1 - y_2$$

$$\Rightarrow 6 + 6y_1 - 6y_2 - 9y_2 + 2y_1 = 0$$

$$\Rightarrow -15y_2 + 8y_1 = -6 \quad \text{--- (c)}$$

On solving (a), (b) & (c)

$$y_0 = y(x_0) = y(0) = \frac{1}{3}$$

$$y_1 = y(x_1) = y(0.5) = \frac{1}{2}$$

$$y_2 = y(x_2) = y(1) = \frac{2}{3}$$

Solution of BVP in Partial differential equation (PDE)

Consider

$$A u_{xx} + 2B u_{xy} + C u_{yy} + F(x, y, u, u_x, u_y) = 0 \quad \text{--- ①}$$

$$\text{Where, } u_x = \frac{\partial u}{\partial x} \quad u_{xx} = \frac{\partial^2 u}{\partial x^2}$$

$$u_{xy} = \frac{\partial^2 u}{\partial y \partial x}, \quad u_y = \frac{\partial u}{\partial y}, \quad u_{yy} = \frac{\partial^2 u}{\partial y^2}$$

Equation ① is

(i) elliptic if $B^2 - AC < 0$

(ii) parabolic if $B^2 - AC = 0$

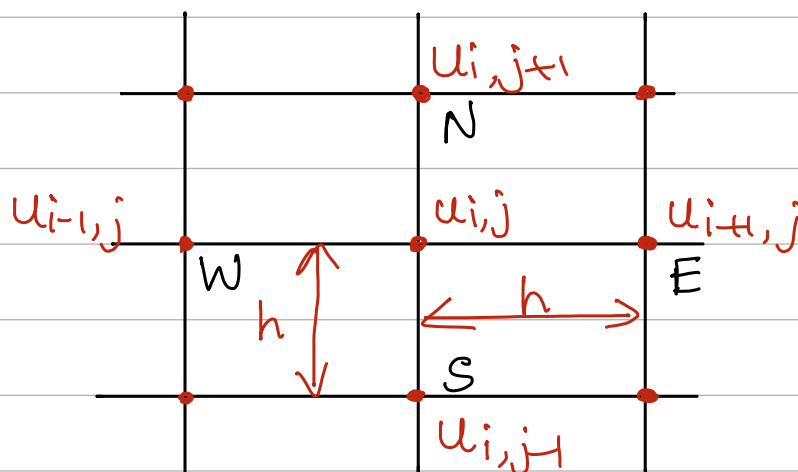
(iii) hyperbolic if $B^2 - AC > 0$

1) Elliptic equations

Laplace's equation:-
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Poisson's equation:-
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = F(x, y)$$

To solve Laplace's and Poisson's equation in a rectangular region, divide the region into small squares by drawing lines parallel to the sides of the rectangle. The points of the intersection of these lines are called mesh points or lattice points or grid points.



Let (x_0, y_0) be an arbitrary interior corner point and $x_i = x_0 + ih$
 $y_j = y_0 + jh$

$$\text{Let } u_{ij} = u(x_i, y_j)$$

$$u_{i+1,j} = u(x_i + h, y_j)$$

$$u_{i-1,j} = u(x_i - h, y_j)$$

By Taylor series expansion,

$$u_{i+1,j} = u_{ij} + h \frac{\partial u_{ij}}{\partial x} + \frac{h^2}{2!} \frac{\partial^2 u_{ij}}{\partial x^2} + \frac{h^3}{3!} \frac{\partial^3 u_{ij}}{\partial x^3} + \dots$$

$$u_{i-1,j} = u_{ij} - h \frac{\partial u_{ij}}{\partial x} + \frac{h^2}{2!} \frac{\partial^2 u_{ij}}{\partial x^2} - \frac{h^3}{3!} \frac{\partial^3 u_{ij}}{\partial x^3} + \dots$$

Adding these two eqns we get,

$$u_{i+1,j} + u_{i-1,j} = 2u_{ij} + \frac{h^2}{2!} \frac{\partial^2 u_{ij}}{\partial x^2} + \frac{h^4}{4!} \frac{\partial^4 u_{ij}}{\partial x^4} + \dots$$

As h is very small, we can neglect $h^4, h^6, \dots$

$$\Rightarrow u_{i+1,j} + u_{i-1,j} = 2u_{ij} + h^2 \frac{\partial^2 u_{ij}}{\partial x^2}$$

$$\Rightarrow \frac{\partial^2 u_{ij}}{\partial x^2} = \frac{u_{i+1,j} - 2u_{ij} + u_{i-1,j}}{h^2}$$

(with error $O(h^2)$)

Similarly

$$\frac{\partial^2 u_{ij}}{\partial y^2} = \frac{u_{i,j+1} - 2u_{ij} - u_{i,j-1}}{h^2}$$

(with error $O(h^2)$)

To solve Laplace eqn

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\text{Set } x = x_i, \quad y = y_j$$

$$\Rightarrow \frac{u_{i+1,j} - 2u_{ij} - u_{i-1,j}}{h^2} + \frac{u_{i,j+1} - 2u_{ij} - u_{i,j-1}}{h^2} = 0$$

$$\Rightarrow u_{ij} = \frac{1}{4} (u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1})$$

$$u_{ij} = \frac{1}{4} (N + E + W + S)$$

(This is to remember easily. N, E, W, S stand for the values of u at the four neighbouring points in north, east, west and south directions respectively)

Similarly to solve Poisson's equation,

$$u_{ij} = \frac{1}{4} (u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - h^2 f_{ij})$$

where $f_{ij} = f(x_i, y_j)$

This formula can be remembered by

$$u_{ij} = \frac{1}{4} (N + E + W + S - h^2 f_{ij}).$$

1. Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ in $0 < x < 1$,

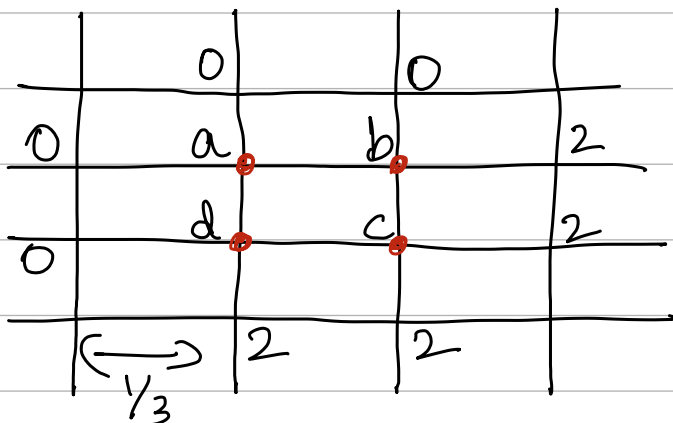
$0 < y < 1$. $u(x, 1) = u(0, y) = 0$,

$u(1, y) = 9(y - y^2)$, $u(x, 0) = 9(x - x^2)$ with $h = \frac{1}{3}$

Solⁿ:- Here $0 < x < 1$, $0 < y < 1$
 $h = \frac{1}{3}$

$x_0 = 0$ $x_1 = \frac{1}{3}$ $x_2 = \frac{2}{3}$ $x_3 = 1$

$y_0 = 0$, $y_1 = \frac{1}{3}$ $y_2 = \frac{2}{3}$ $y_3 = 1$



Boundary conditions

$$u(x, 1) = u(0, y) = 0, \quad u(1, y) = 9(y - y^2)$$

$$u(x, 0) = 9(x - x^2)$$

Left edge of rectangle is $x = 0$

Given $u(0, y) = 0 \quad \therefore$ Points at left edge are zeros.

Top edge is $y = 1$. Given. $u(x, 1) = 0$
 $\Rightarrow$ Points at top edge are zeros.

Right edge of rectangle is $x = 1$

$$\text{Given } u(1, y) = 9(y - y^2)$$

$$\text{Points are } u(1, \frac{1}{3}) = 9(\frac{1}{3} - \frac{1}{3}^2) = 2$$

$$u(1, \frac{2}{3}) = 9(\frac{2}{3} - \frac{4}{9}) = 2$$

Bottom edge of rectangle is $y = 0$

$$\text{Given } u(x, 0) = 9(x - x^2)$$

$$\text{Points are } u(\frac{1}{3}, 0) = 2$$

$$u(\frac{2}{3}, 0) = 2$$

The unknown values of u at the grid points are a, b, c, d .

$\therefore$ From the grid

$$a = \frac{b+d}{4}$$

$$b = \frac{a+c+2}{4}$$

$$c = \frac{b+d+4}{4}$$

$$d = \frac{a+c+2}{4}$$

Here $b=d$

$$\Rightarrow a = \frac{2b}{4} = \frac{b}{2}$$

$$b = \frac{a+c+2}{4}$$

$$c = \frac{2b+4}{4} = \frac{b+2}{2}$$

$$\Rightarrow 2a - b = 0$$

$$a - 4b + c = -2$$

$$b - 2c = -2$$

On solving

$$a = \frac{1}{2}, \quad b = 1, \quad c = \frac{3}{2}, \quad d = 1$$

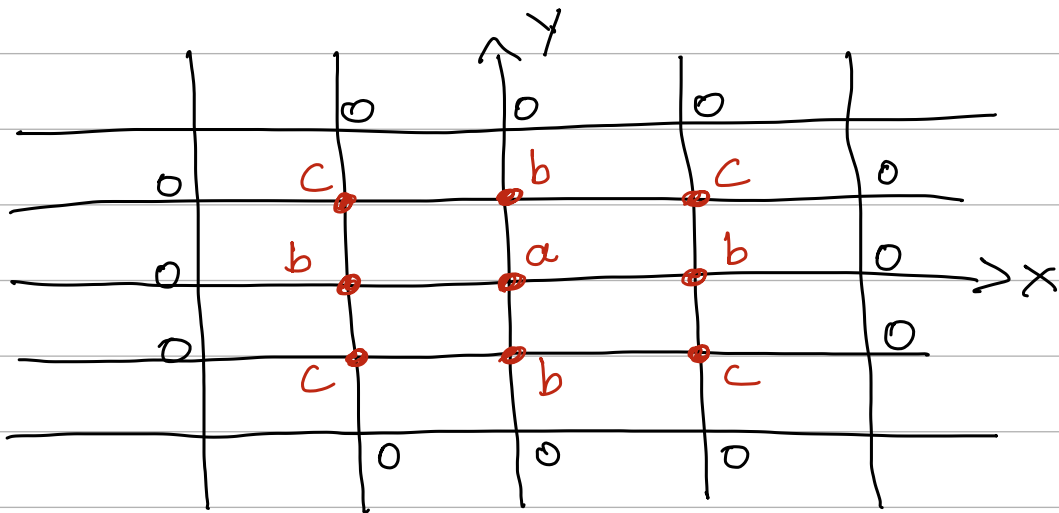
	0		0	
0	a	b		2
0	d	c		2
	2		2	

2) With step size $h = \frac{1}{2}$, solve

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -1, \quad |x| < 1, \quad |y| < 1$$

$$u(\pm 1, y) = u(x, \pm 1) = 0$$

Solⁿ:- $-1 < x < 1, \quad -1 < y < 1$



$$-1, -0.5, 0, 0.5, 1 \rightarrow x$$

$$-1, -0.5, 0, 0.5, 1 \rightarrow y$$

Given $u(1, y) = 0 \Rightarrow x = 1$ Right edge 0

$u(-1, y) = 0 \Rightarrow x = -1$, left edge 0

$u(x, 1) = 0 \Rightarrow y = 1$, top edge 0

$u(x, -1) = 0 \Rightarrow y = -1$, bottom edge 0

Because of symmetry, we have 3 unknowns.

$$u_{ij} = \frac{1}{4} (N + W + S + E - h^2 f_{ij})$$

$$\Rightarrow a = \frac{1}{4} (4b + \frac{1}{4})$$

$$b = \frac{1}{4} (a + 2c + \frac{1}{4})$$

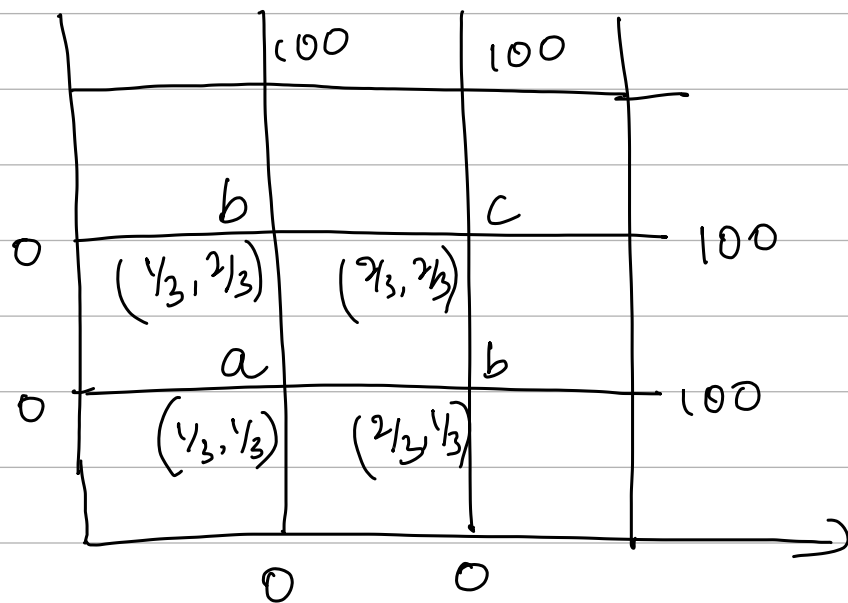
$$c = \frac{1}{4} (2b + \frac{1}{4})$$

3) With $h = \frac{1}{3}$, solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -81xy$,
 $0 < x < 1, 0 < y < 1$

$$u(0, y) = u(x, 0) = 0$$

$$u(1, y) = u(x, 1) = 100$$

Soln.:-



$$f(x, y) = -81xy$$

$$a = \frac{1}{4} (2b + \frac{1}{9} \times 81 \times \frac{1}{3} \times \frac{1}{3}) = \frac{1}{4} (2b + 1)$$

$$\Rightarrow a = \frac{1}{4} (2b + 1)$$

$$b = \frac{a + c + 100 + \frac{1}{4}a \times 81 \times \frac{2}{3} \times \frac{1}{3}}{4}$$

$$= \frac{a + c + 100 + 2}{4}$$

$$\Rightarrow b = \frac{a + c + 102}{4}$$

$$c = \frac{2b + 200 + \frac{1}{4}a \times 81 \times \frac{2}{3} \times \frac{1}{3}}{4}$$

$$\Rightarrow c = \frac{2b + 204}{4}$$

$$\Rightarrow a = 26, \quad b = 51, \quad c = 76$$

Parabolic equation

Heat equation.

$$\frac{\partial u}{\partial t} = c \frac{\partial^2 u}{\partial x^2}, \quad a < x < b, \quad t > 0 \quad \text{--- ①}$$

$$u(x, 0) = f(x) \quad \text{--- ②}$$

$$u(a, t) = \phi(t), \quad u(b, t) = \psi(t), \quad t > 0 \quad \text{--- ③}$$

Divide $[a, b]$ into n equal parts
each of length h .

$$\text{i.e., } h = \frac{b-a}{n}.$$

$$\text{Let } x_i = a + ih, \quad i = 0, 1, 2, \dots, n$$

Let k be the time interval size and
let $t_j = jk$

$$\text{Let } u(x_i, t_j) = u_{ij}$$

$$u_{i,j+1} = u_{ij} + k \frac{\partial u_{ij}}{\partial t} + \frac{k^2}{2!} \frac{\partial^2 u_{ij}}{\partial t^2} + \dots$$

Neglecting higher order terms $k^2, k^4, \dots$

$$\Rightarrow \frac{\partial u_{ij}}{\partial t} = \frac{u_{i,j+1} - u_{ij}}{k} \quad \text{--- ④}$$

Also

$$\frac{\partial^2 u_{ij}}{\partial t^2} = \frac{1}{h^2} (u_{i+1,j} - 2u_{ij} + u_{i-1,j}) \quad \text{--- (b)}$$

Put (a) & (b) in (1)

$$\Rightarrow u_{i,j+1} = u_{ij} + \frac{kc}{h^2} (u_{i+1,j} - 2u_{ij} + u_{i-1,j})$$

$$\text{or } u_{i,j+1} = \lambda (u_{i+1,j} + u_{i-1,j}) + (1-2\lambda)u_{ij}$$

$$\text{where } \lambda = \frac{kc}{h^2}$$

If $\lambda = \frac{1}{2}$, then

$$u_{i,j+1} = \frac{(u_{i+1,j} + u_{i-1,j})}{2} \quad \text{--- (c)}$$

Eqn (c) is used to find $u(x, y)$ at time $t = t_{j+1}$ from the values of u at $t = t_j$

From (2), we have

$$u(x, 0) = u_{i0} = f_i$$

From (3), we have

$$u_{0j} = \phi_j \text{ and } u_{nj} = \psi_j$$

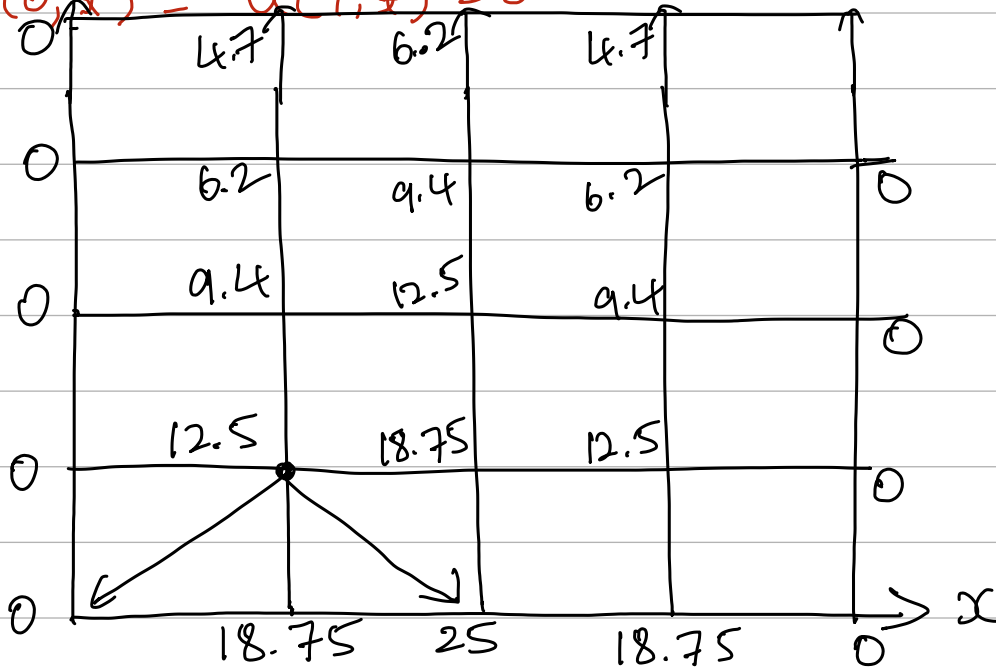
$$f_i \rightarrow f(x_i), \quad \phi_j = \phi(t_j), \quad \psi_j = \psi(t_j)$$

1. Compute u for four time steps with $h=0.25$

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < 1, \quad t > 0$$

$$u(x, 0) = 100(x - x^2)$$

$$u(0, t) = u(1, t) = 0$$



Given $u(x, 0) = 100(x - x^2)$

(Put $x = 1/4$) $u_{10} = 100 \left(\frac{1}{4} - \frac{1}{16} \right) = 18.75$

(Put $x = 1/2$) $u_{20} = 100 \left(\frac{1}{2} - \frac{1}{4} \right) = 25$

(Put $x = 3/4$) $u_{30} = 100 \left(\frac{3}{4} - \frac{9}{16} \right) = 18.75$

$$u_{i,j+1} = \frac{u_{i+1,j} + u_{i-1,j}}{2}$$

$$u_{11} = \frac{25 + 0}{2} = 12.5$$

$$u_{12} = \frac{18.75 + 18.75}{2} = 18.75$$

$$u_{13} = \frac{25 + 0}{2} = 12.5$$

$$\text{Here } k = \frac{h^2}{2c} = \frac{\left(\frac{1}{4}\right)^2}{2 \times 1} = \frac{1}{\underline{\underline{32}}}$$

$k \rightarrow$ time step

Hyperbolic equation

Wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad a < x < b, \quad t > 0 \quad \text{--- (1)}$$

$$u(x, 0) = f(x), \quad \frac{\partial u}{\partial t}(x, 0) = g(x) \quad \text{--- (2)}$$

$$u(a, t) = \phi(x), \quad u(b, t) = \psi(x) \quad \text{--- (3)}$$

Divide $[a, b]$ into n equal parts
each of length h .

$$\text{i.e., } h = \frac{b-a}{n}.$$

$$\text{Let } x_i = a + ih, \quad i = 0, 1, 2, \dots, n \quad \text{--- (4)}$$

Let k be the time interval size and
let $t_j = jk$ (5)

$$\text{Let } u(x_i, t_j) = u_{ij}$$

We know,

$$\frac{\partial^2 u}{\partial x^2} = \frac{u_{i+1,j} - 2u_{ij} + u_{i-1,j}}{h^2} \quad \text{--- (6)}$$

Similarly

$$\frac{\partial^2 u}{\partial t^2} = \frac{u_{i,j+1} - 2u_{ij} + u_{i,j-1}}{k^2} \quad \text{--- (7)}$$

Using (6) & (7) in (1)

$$\frac{u_{i,j+1} - 2u_{ij} + u_{i,j-1}}{k^2} = \frac{c^2}{h^2} (u_{i+1,j} - 2u_{ij} + u_{i-1,j}) \quad \text{--- (8)}$$

$$\text{If } k = \frac{h}{c}$$

(8) $\Rightarrow$

$$u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1} \quad \text{--- (9)}$$

This is the finite difference scheme for wave equation.

If we know u at two consecutive time steps t_{j-1} , t_j then (9) gives u at time $t = t_{j+1}$

The values u_{i0} are obtained from eqn (2)

$$u_{i0} = f_i \quad \text{--- (10)}$$

To get u_{i1} we have from (9)

$$\begin{aligned} u_{i1} &= u_{i+1,0} + u_{i-1,0} - u_{i,-1} \\ &= f_{i+1} + f_{i-1} - u_{i0} + k \frac{\partial u_{i0}}{\partial t} - \frac{k^2}{2!} \frac{\partial^2 u_{i0}}{\partial t^2} + \dots \end{aligned}$$

$$= f_{i+1} + f_{i-1} - f_i + kg_i - \frac{h^2}{2c^2} \frac{\partial^2 u_{i0}}{\partial t^2} \dots$$

$$= f_{i+1} + f_{i-1} - f_i + kg_i - \frac{h^2}{2} \frac{\partial^2 u_{i0}}{\partial x^2}$$

$$= f_{i+1} + f_{i-1} - f_i + kg_i - \frac{h^2}{2} f_i'' \quad (\text{from (1)})$$

$$= f_{i+1} + f_{i-1} - f_i + kg_i - \frac{1}{2} (f_{i+1} - 2f_i + f_{i-1})$$

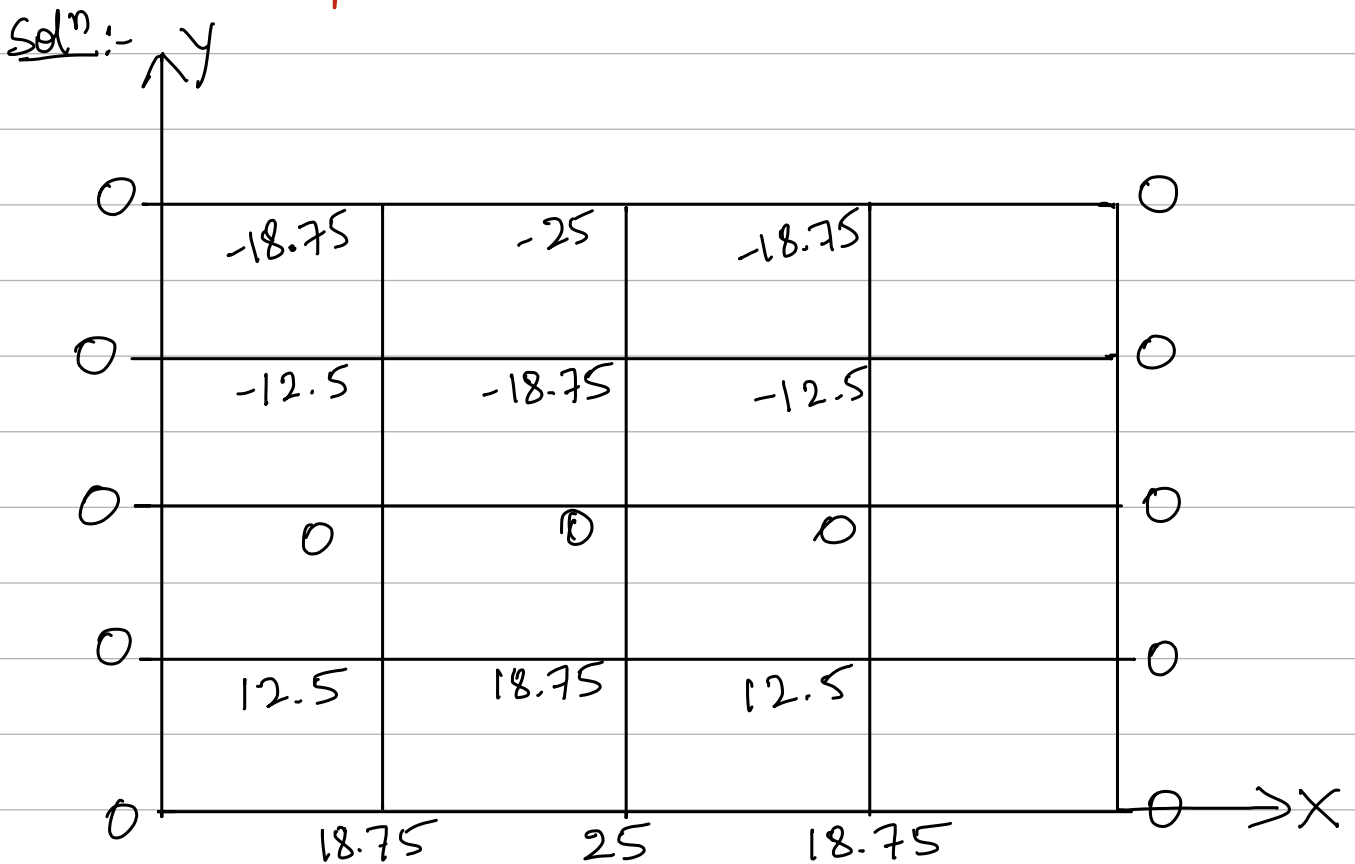
$$\text{or } u_{i1} = \frac{1}{2} (f_{i+1} + f_{i-1}) + kg_i \quad \text{--- (11)}$$

Egn (10) & (11) give values of u at zeroth and first time step. Then (9) gives u at second time step and so on.

1. Solve $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$, $0 < x < 1$, $t > 0$

$u(x, 0) = 100(x - x^2)$, $\frac{\partial u}{\partial t}(x, 0) = 0$

$u(0, t) = u(1, t) = 0$ with $h = 0.25$ upto 4 steps



Initial condition,

$u(x, 0) = 100(x - x^2)$

$u_{10} = 100\left(\frac{1}{4} - \frac{1}{16}\right) = 18.75 \quad (x = \frac{1}{4})$

$u_{20} = 100\left(\frac{1}{2} - \frac{1}{4}\right) = 25 \quad (x = \frac{1}{2})$

$u_{30} = 100\left(\frac{3}{4} - \frac{9}{16}\right) = 18.75 \quad (x = \frac{3}{4})$

Using $u_{i1} = \frac{1}{2} (f_{i+1} + f_{i-1}) + kg_i$

Here $g_i = \frac{\partial u}{\partial x}(x, 0) = 0$

$$\Rightarrow u_{i1} = \frac{1}{2} (f_{i+1} + f_{i-1})$$
$$= \frac{1}{2} \{ u_{i+1} + u_{i-1} \}$$

$$u_{11} = \frac{1}{2} (u_{20} + u_{00}) = \frac{1}{2} (25) = 12.5$$

$$u_{21} = \frac{1}{2} (u_{30} + u_{10}) = \frac{18.75 + 18.75}{2} = 18.75$$

$$u_{31} = \frac{1}{2} (u_{40} + u_{20}) = \frac{1}{2} (0 + 25) = 12.5$$

Consider

$$u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1}$$

$$u_{12} = u_{21} + u_{01} - u_{10}$$
$$= 18.75 + 0 - 18.75 = 0$$

$$u_{22} = u_{31} + u_{11} - u_{20}$$
$$= 12.5 + 12.5 - 25 = 0$$

$$u_{32} = u_{41} + u_{21} - u_{30}$$
$$= 0 + 18.75 - 18.75 = 0$$

$$u_{13} = u_{22} + u_{02} - u_{11}$$

$$= 0 + 0 - 12.5 = -12.5$$

$$u_{23} = u_{32} + u_{12} - u_{21}$$

$$= 0 + 0 - 18.75 = -18.75$$

$$u_{33} = u_{42} + u_{22} - u_{31}$$

$$= 0 + 0 - 12.5 = -12.5$$

$$u_{14} = u_{23} + u_{03} - u_{21}$$

$$= -18.75 + 0 - 0 = -18.75$$

$$u_{24} = u_{33} + u_{13} - u_{22}$$

$$= -12.5 - 12.5 - 0 = -25$$

$$u_{34} = u_{43} + u_{23} - u_{32}$$

$$= 0 + (-18.75) - 0 = -18.75$$